

RAG Assistant — Chat With Your Business Documents

Grounded answers from a business's own docs — every fact cited, and an honest handoff when the answer isn't there.

THE PROBLEM

A generic chatbot answers from its training data — a blurry average of the internet — so it confidently invents prices, policies and details that are wrong for **your** business. In front of real customers, a made-up cancellation fee or a wrong opening time is a liability, not a convenience.

WHAT IT DOES

- Answers customer questions using **only** the business's own documents (FAQs, pricing, policies).
- Cites the exact source passage behind every answer, so it's verifiable.
- Refuses to guess: when the answer isn't in the docs it says *"I'll pass this to the team"* instead of hallucinating.
- Update the documents and the answers update instantly — no retraining.

HOW IT WORKS

documents → section-aware chunking → embeddings → pgvector store → semantic retrieval → grounded generation with citations

The model reasons; the retrieved passages provide the facts. Generation is constrained to those passages, which is what makes it safe to deploy customer-facing.

BUILT & SHIPPED

- Live, deployed app + public code repository.
- Documented real engineering problems hit and fixed: a retired embedding model, an approximate index silently returning zero results, and naive chunking retrieving the wrong section.
- Fixes measured with a before/after retrieval evaluation.

WHY IT MATTERS

- "Grounded or silent" — never invents a price or policy.
- Every business has documents customers ask about; this turns them into a reliable self-serve assistant.
- The same pattern powers document-grounded internal tools and support automation.

STACK

Python · FastAPI · Postgres + pgvector · Gemini embeddings & generation · Docker · deployed on Render.